

Narasinga Ujjini

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Focused and driven future engineer with expertise in microcontrollers, sensors, programming languages, and machine learning tools. Known for commitment, initiative, and curiosity.

- FCC HAM Radio Certified - Technician Class License - Callsign KO6YO
- AWS Certified Cloud Practitioner
- Scouting America - Eagle Scout
- Expertise in Java, Python, Godot, Arduino IDE, EagleCAD, Fusion360, Advanced Excel Techniques, RDWorksV8, HTML
- Expertise in Arduino Uno, Intel Edison, Raspberry Pi, Metro M0, ESP32, RoboRio FPGA
- Expertise with various sensors: HCSR04 ultrasonic sensor, DHT22/DHT11 Temperature/Humidity Sensors, Servos/Motors

Work Experience (References Available)

- **San Diego County Fair - Shuttle Ambassador in Public Safety Department** (Jun. '25 - Jul. '25; Jun. '26 - Jul. '26)
 - Used leadership and crowd-management skills to handle ~2,000 daily Fair guests
 - Optimized flow of guest traffic between the fair and satellite parking locations using real-time bus location
- **Inclusive Tutoring - Business Development, Customer Experience, Content Curating** (Jun. '24 - Present)
 - AI-driven educational business providing tutoring and support resources to special needs students
 - Focused on students' individual needs with aesthetic, welcoming website
- **San Diego Unified High School District Food Service - Digitized Data Management and Efficiency** (Oct. '25 - Feb. '25)
 - Saved ~10hr/wk of labor (~\$330/wk) by digitizing inventory management, creating more time to serve customers
 - Used Google Drive OCR tool to scan delivery list and upload to Google Sheets
- **UCSD ECE Student Internship Preparation Program (Prof. Karcher Morris)** (Jul. '24 - Aug. '24)
 - Won the product design competition - Created a fire safety device to alert parents if children came too close to a fireplace. Components included an IR sensor, Adafruit Metro M0 board, buzzer, and LEDs.
 - Learned PCB Design in EAGLE, and electrical circuit design with Adafruit Metro M0
 - Won the 'Best Overall CAD Model' competition with a Brake Rotor/Caliper assembly in Onshape

Education

Torrey Pines High School

- Graduated in May 2026 with Honors - 4.4 GPA
- Excelled in AP Courses: Physics 1, 2, C; Calculus AB; Computer Science A, Statistics courses (4+ score on all exams)
- AP Scholar™ with Distinction - Scored 3 or higher on at least 5 AP Exams
- National Merit Commended Student - Scored in the 99th percentile of PSAT scores
- CollegeBoard™ National Recognition Program - Scored in the top 10% of Torrey Pines High School's PSAT scores, and earned a 3 or higher on at least 5 AP exams
- Wrestled for 4 years, placed at local varsity tournaments

MiraCosta Community College - Math & Sciences Associate Degree

- Took college courses while still in High School for more technical depth
- Calculus III, Linear Algebra, and Differential Equations, Java, Computer Architecture
- CS 112 Capstone Project - Created an Instrument Practice tracking tool with Java, JavaFX, Maven, and IntelliJ

Projects

Engineering Projects (2026)

- Python Tuner & Fourier Transform Proof-of-Concept / [Link to Portfolio Page](#)
 - Used NumPy and PyQtGraph libraries for audio processing and visualization.
 - Used a Fast Fourier Transform (FFT) to identify the dominant frequency and convert it to a musical note

Circuit Design/Electronics (2025-26)

- Used the Google Coral Tensor Processing Unit (TPU) AI USB Accelerator for **computer vision and image detection** to detect if a stove is on or not
 - Trained an image classification model (MobileNet) using Google Teachable Machine and Edge Impulse
 - Combined the above model with a Python script to alert the user if their stove has been left on
 - Used a pre-trained 'PoseNet' model to identify human poses (smiles, jumping jacks, etc.)
- Used Behringer and Neural DSP Interfaces with Python modulation libraries to create a unique electric guitar signal chain, such as noise gates, distortion, and delay

Amazon Alexa Skills Challenge (2022)

- Won both the individual and group competitions - built Alexa 'skills' (apps) for Amazon Alexa users
- **Individual Challenge:** Developed 'American National Park Campgrounds' application for Amazon Alexa via Voiceflow and nps.gov API to identify availability of campgrounds in American National Parks
 - Used Google Sheets API to manage data and filter for user-appropriate information
- **Group Challenge:** Created School Lunch application for via voiceflow that enabled parents to set a schedule for children's school lunches
- [More Projects on my Website](#)